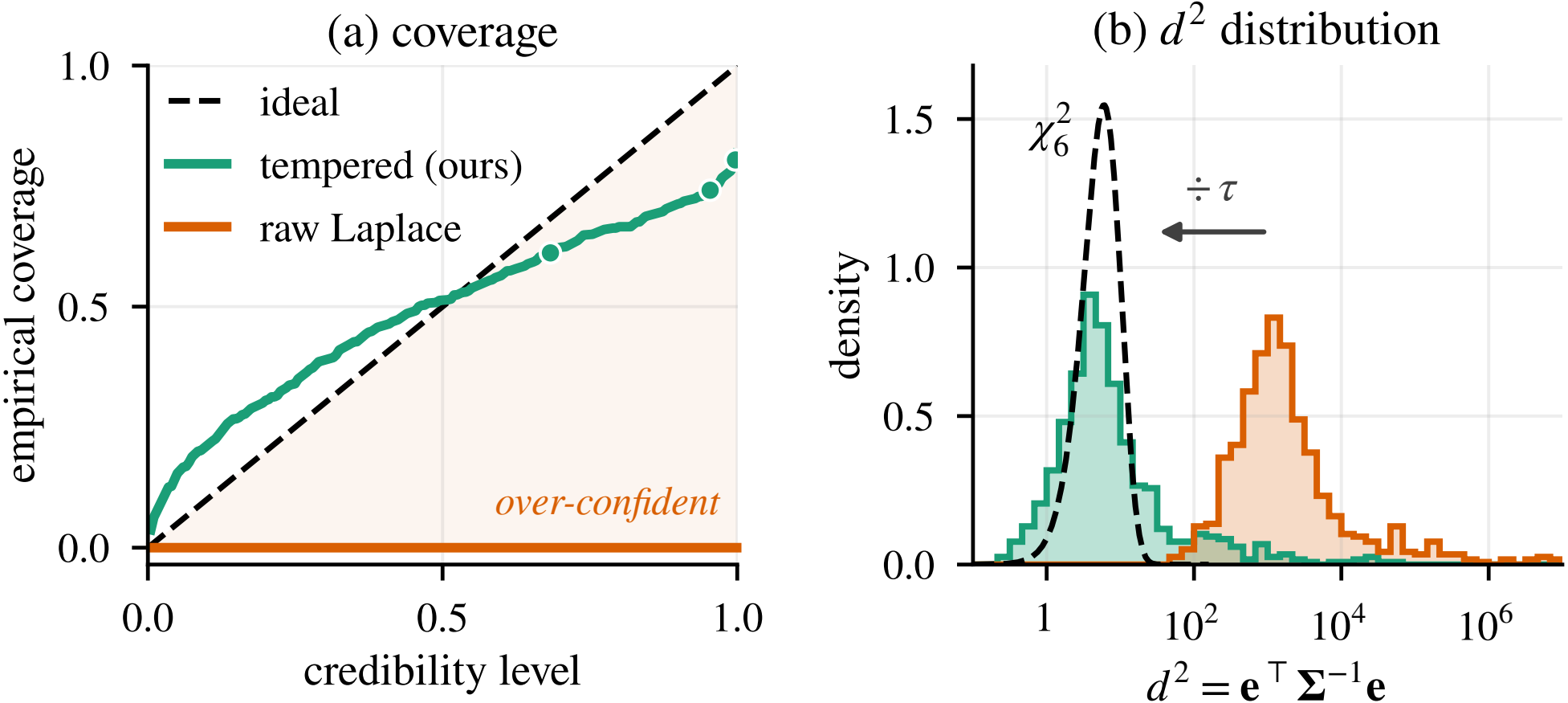
(a) coverage
ideal
tempered (ours)
raw Laplace
over-confident
empirical coverage
credibility level
1.0
0.5
0.0
0.0
0.5
1.0
(b) $d^2$ distribution
$\chi^2_6$
$\div \tau$
density
1.5
1.0
0.5
0.0
1
$10^2$
$10^4$
$10^6$
$d^2 = \mathbf{e}^\top \boldsymbol{\Sigma}^{-1} \mathbf{e}$